

A Unified Structural Counterfactual Framework for Domain-General Reasoning Under Uncertainty

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Abstract—This paper presents a unified structural counterfactual framework for domain-general causal reasoning under uncertainty. While existing approaches in causal inference and counterfactual analysis often operate in isolation or rely on domain-specific assumptions, they lack an integrated pipeline capable of handling heterogeneous datasets with consistent reasoning guarantees. To address this limitation, we propose a multi-phase framework that systematically combines data understanding, causal graph construction, effect estimation, counterfactual simulation, uncertainty quantification, and cross-domain generalization. The framework leverages structural causal models for effect identification and integrates machine learning-based counterfactual estimation to capture individual-level variations. Uncertainty is explicitly quantified using bootstrap-based estimation, and robustness is validated through refutation techniques. The proposed approach is evaluated across diverse domains, including education, healthcare, and marketing datasets, demonstrating its ability to generalize causal insights while maintaining consistency between estimated and simulated effects. Experimental results show that the framework produces stable causal estimates with low variance in most domains, while highlighting domain-specific variability that inform reliability. The proposed system provides a scalable and interpretable solution for real-world decision-making scenarios where both causality and uncertainty play critical roles.

Index Terms—Causal Inference, Counterfactual Analysis, DoWhy, Machine Learning, Robustness, Generalization

I. INTRODUCTION

Causal reasoning has become an essential component of modern data-driven systems, particularly in domains where understanding the impact of interventions is more critical than predictive accuracy alone. Traditional machine learning methods primarily rely on correlations and statistical associations, which limits their applicability in decision-making scenarios such as education, healthcare, and marketing. These limitations have motivated the development of causal inference frameworks that explicitly model cause–effect relationships and enable reasoning beyond observed data [1]–[3].

Structural causal models (SCMs) provide a formal foundation for representing causal mechanisms and allow for the identification and estimation of causal effects from observational data [1], [3]. In addition, counterfactual analysis enables reasoning about hypothetical scenarios by answering “what-if” questions, making it a powerful tool for both policy evaluation and individualized decision-making. Despite these advancements, existing approaches often treat causal estimation and

counterfactual reasoning as separate processes, resulting in fragmented pipelines that lack consistency and interpretability.

Recent developments in causal inference have been supported by practical frameworks such as DoWhy, which provides an end-to-end interface for modeling, identification, estimation, and refutation of causal effects [4]. However, most implementations remain domain-specific and require manual configuration of causal structures, limiting their scalability across heterogeneous datasets. Furthermore, while machine learning models such as Random Forests are effective in capturing complex nonlinear relationships and are widely used for predictive and counterfactual modeling [5], [6], they are often not tightly integrated with causal reasoning pipelines.

Another significant challenge lies in the ability of causal systems to generalize across domains. Real-world datasets often exhibit distributional differences, making it difficult for a single model to maintain consistent performance. Domain generalization techniques aim to address this issue by learning invariant representations that remain stable across varying environments [7]. However, these approaches are rarely combined with structural causal frameworks, resulting in a gap between theoretical robustness and practical applicability.

In addition to generalization, uncertainty quantification is crucial for interpreting causal estimates reliably. While causal effects can be computed using statistical or machine learning approaches, the variability associated with these estimates is often overlooked. Bootstrap methods provide an effective mechanism for estimating uncertainty and constructing confidence intervals, thereby improving the reliability of causal conclusions [8].

To address these challenges, this paper proposes a unified structural counterfactual framework for domain-general reasoning under uncertainty. The proposed framework integrates multiple stages, including data understanding, causal graph construction, effect identification and estimation using structural models, counterfactual simulation using machine learning techniques, uncertainty quantification through bootstrap methods, and robustness validation using refutation strategies. Unlike conventional approaches, the framework is designed to operate consistently across multiple domains without requiring extensive domain-specific customization.

The proposed approach is evaluated on diverse real-world datasets spanning different domains, including the Student Performance dataset [9], [10], the Pima Indians Diabetes dataset [11], and the MineThatData email marketing dataset

introduced by Kevin Hillstrom [12]. These datasets represent educational, healthcare, and marketing contexts, respectively, enabling a comprehensive assessment of the framework’s ability to generalize causal reasoning across heterogeneous environments.

Experimental results demonstrate that the proposed framework produces stable causal effect estimates while maintaining consistency between structural estimation and counterfactual predictions. Furthermore, the integration of uncertainty quantification provides additional insights into the reliability of results across domains. By combining causal inference, counterfactual reasoning, domain generalization, and uncertainty analysis within a unified pipeline, this work contributes a scalable and interpretable framework for real-world decision-making applications.

The remainder of this paper is organized as follows. Section II reviews related work in causal inference and domain generalization. Section III presents the proposed methodology. Section IV describes the experimental setup. Section V discusses the results, followed by robustness analysis in Section VI. Limitations are presented in Section VII, and conclusions are drawn in Section VIII.

II. RELATED WORK

Causal inference has been extensively studied as a framework for understanding cause–effect relationships beyond statistical correlations. Foundational work by Pearl introduced structural causal models (SCMs), providing a formal framework for representing causal mechanisms and enabling reasoning through interventions and counterfactuals [1]. Rubin’s potential outcomes framework further established a statistical perspective for estimating causal effects under treatment-control settings [2]. These approaches have been unified and extended in subsequent works, including the formulation of causal discovery and learning algorithms that bridge theory and practical implementation [3].

In recent years, practical tools have been developed to operationalize causal inference in real-world applications. The DoWhy framework offers an end-to-end pipeline that integrates causal modeling, effect identification, estimation, and refutation within a unified interface [4]. While such tools significantly simplify implementation, they often rely on predefined causal graphs and are typically applied within single-domain settings, limiting their adaptability to diverse datasets.

Machine learning models have also been widely incorporated into causal analysis pipelines, particularly for modeling complex relationships and supporting counterfactual estimation. Ensemble methods such as Random Forests are known for their robustness and ability to capture nonlinear interactions, making them suitable for predictive and simulation-based tasks [5]. In addition, statistical learning frameworks provide a strong foundation for integrating data-driven modeling techniques within broader analytical pipelines [6]. However, these methods primarily focus on prediction and are not inherently designed for causal reasoning, leading to a disconnect between predictive performance and causal interpretability.

Another important line of research focuses on uncertainty quantification in statistical and machine learning models. Bootstrap methods provide a practical approach for estimating variability and constructing confidence intervals, allowing for more reliable interpretation of estimated effects [8]. Despite their effectiveness, uncertainty estimation is often treated as a separate component rather than being integrated directly into causal reasoning pipelines.

Domain generalization has emerged as a key area for addressing distributional shifts across datasets. Approaches such as invariant feature learning aim to extract representations that remain stable across different environments, thereby improving model generalization [7]. While these techniques enhance robustness in predictive settings, they are rarely combined with structural causal models and counterfactual reasoning in a unified framework.

Existing studies have also explored causal and predictive modeling in specific application domains. For instance, the Student Performance dataset has been used to analyze educational outcomes and identify factors influencing academic performance [9], [10]. Similarly, healthcare datasets such as the Pima Indians Diabetes dataset are commonly used for studying risk factors and predictive modeling in medical contexts [11]. In the marketing domain, the MineThatData email analytics dataset introduced by Kevin Hillstrom provides a benchmark for evaluating customer response to targeted interventions [12]. However, these studies are typically conducted in isolation and do not emphasize cross-domain causal generalization.

Despite significant progress across these areas, a gap remains in integrating causal inference, counterfactual reasoning, uncertainty quantification, and domain generalization into a single cohesive framework. Most existing approaches address these components independently, resulting in fragmented pipelines that lack consistency and scalability.

In contrast, the proposed work introduces a unified structural counterfactual framework that systematically integrates these components into a multi-stage pipeline. By combining causal modeling, counterfactual simulation, bootstrap-based uncertainty estimation, and cross-domain evaluation, the framework addresses the limitations of existing approaches and provides a consistent methodology for domain-general causal reasoning.

III. METHODOLOGY

A. Overall Framework Architecture

The proposed framework is designed as a unified multi-stage pipeline for structural causal reasoning, counterfactual simulation, uncertainty quantification, and domain-general evaluation under heterogeneous environments. Unlike conventional approaches that separately address causal estimation, counterfactual analysis, and robustness validation, the proposed system integrates these components within a single interpretable workflow. The overall architecture of the framework is illustrated in Fig. 1.

The framework begins with heterogeneous input datasets collected from multiple domains, including education, healthcare, and marketing. Data understanding and schema definition are first performed to identify treatment variables, outcome

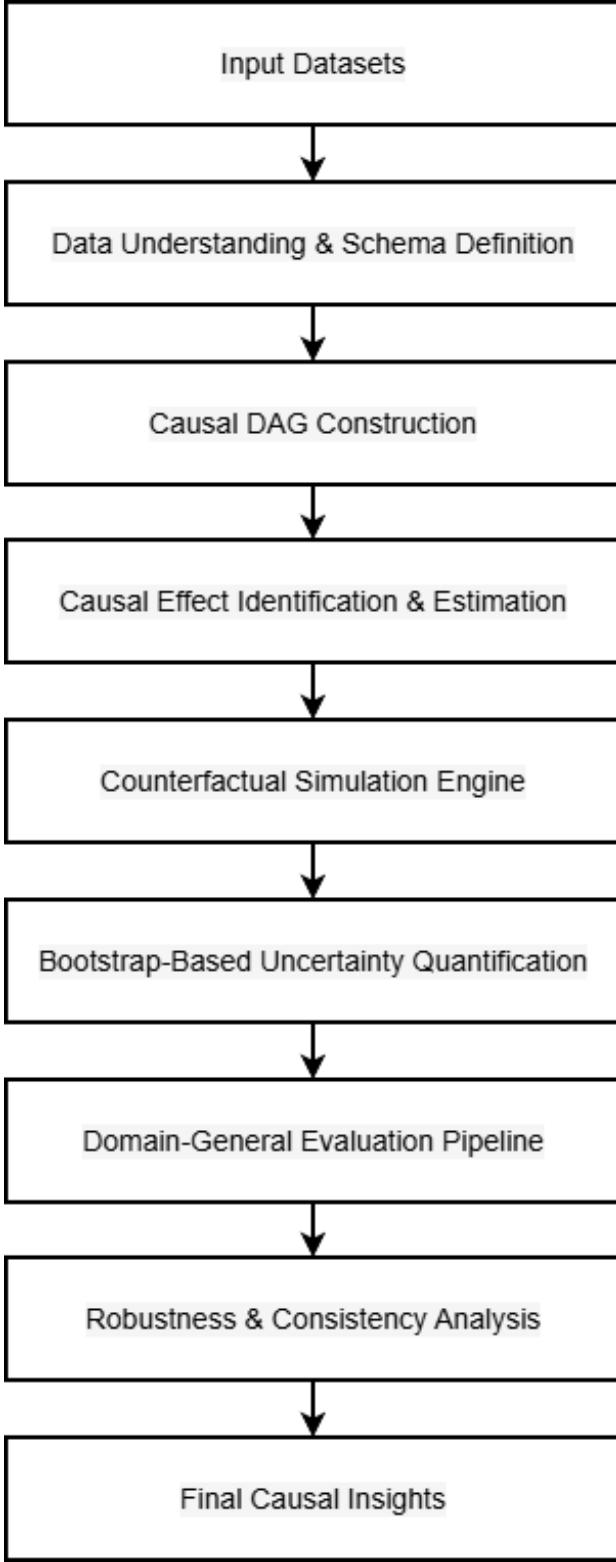


Fig. 1. Overall architecture of the proposed unified structural counterfactual framework for domain-general causal reasoning under uncertainty.

variables, confounding factors, and relevant predictive features. A directed acyclic graph (DAG) is then constructed to represent causal dependencies between variables. Based on the generated causal structure, causal effects are identified and

estimated using structural causal modeling techniques. Counterfactual reasoning is subsequently performed using a machine learning-based simulation engine to estimate individual-level intervention effects. To improve reliability, bootstrap-based uncertainty quantification is incorporated to estimate statistical variability and confidence intervals. The framework is further extended to a domain-general evaluation pipeline that enables causal reasoning across heterogeneous datasets while maintaining consistency between causal estimation and counterfactual simulation. Finally, robustness and consistency analysis are performed to validate the reliability of estimated causal effects under varying conditions.

B. Data Understanding and Schema Definition

The first stage of the proposed framework focuses on data understanding and schema definition. In this stage, datasets are loaded and analyzed to identify relevant variables required for causal modeling and counterfactual analysis. Statistical summaries, feature inspection, and missing value analysis are performed to understand the structure and quality of the data prior to causal processing.

A schema configuration is then defined to standardize the causal pipeline across datasets. The schema contains four primary components: treatment variables, outcome variables, confounding variables, and predictive features. For the educational dataset, study time is selected as the treatment variable, while final student performance is considered the outcome variable. Confounding variables include factors such as absences, health condition, and previous failures, since these variables may simultaneously influence both treatment and outcome.

The schema representation can be formally expressed as:

$$S = \{T, Y, C, F\}$$

where T denotes the treatment variable, Y denotes the outcome variable, C represents the set of confounding variables, and F represents the set of predictive features used for counterfactual estimation.

The defined schema is serialized and stored for reuse in subsequent stages of the framework, ensuring consistency between causal graph construction, effect estimation, and counterfactual simulation.

C. Causal DAG Construction

After schema definition, a directed acyclic graph (DAG) is constructed to represent structural causal dependencies among variables. The proposed framework utilizes a programmatic DAG generation process in which treatment, outcome, and confounding variables are connected through directed causal edges.

The causal graph can be represented as:

$$G = (V, E)$$

where V denotes the set of variables and E denotes the set of directed causal edges between variables.

The framework establishes a direct causal relationship between treatment and outcome variables:

$$T \rightarrow Y$$

In addition, each confounding variable is connected to both treatment and outcome variables:

$$C_i \rightarrow T$$

$$C_i \rightarrow Y$$

where C_i represents the i^{th} confounding variable.

The generated DAG structure is subsequently utilized within the structural causal modeling stage for effect identification and estimation. By explicitly modeling confounding relationships, the framework improves interpretability and supports causal reasoning under observational settings.

D. Causal Effect Identification and Estimation

Following causal graph construction, causal effect identification and estimation are performed using the DoWhy causal inference framework. The generated DAG is integrated with observational data to construct a structural causal model capable of identifying causal relationships between treatment and outcome variables.

The framework first identifies a valid estimand based on the backdoor adjustment criterion and then estimates the causal effect using linear regression-based adjustment. The average treatment effect (ATE) is computed as:

$$ATE = \mathbb{E}[Y(1) - Y(0)]$$

where $Y(1)$ represents the potential outcome under intervention and $Y(0)$ represents the potential outcome without intervention.

The estimation process enables the framework to quantify the expected causal influence of treatment variables on outcomes while accounting for confounding effects. The identified causal estimand and estimated treatment effects are stored as reusable artifacts for subsequent counterfactual and robustness analysis stages.

E. Counterfactual Simulation Engine

To support individualized causal reasoning, the proposed framework incorporates a counterfactual simulation engine based on the Random Forest Regressor model. Unlike global causal estimation methods that compute population-level effects, the counterfactual engine estimates instance-specific intervention outcomes under hypothetical scenarios.

The simulation process begins by selecting an observational sample and generating a counterfactual variant through intervention on the treatment variable. The predictive model is then used to estimate both original and counterfactual outcomes.

The counterfactual effect is computed as:

$$\Delta_{cf} = \hat{Y}_{cf} - \hat{Y}_{orig}$$

where $\hat{Y}_{cf}$ denotes the predicted counterfactual outcome and $\hat{Y}_{orig}$ denotes the predicted observational outcome.

The framework iteratively applies this process across all observational instances to estimate individual-level treatment effects. The average of these simulated effects provides an aggregated estimate of intervention impact across the dataset. The use of Random Forest regression enables the framework to capture nonlinear relationships and feature interactions during counterfactual estimation.

F. Bootstrap-Based Uncertainty Quantification

To improve reliability and interpretability, the proposed framework incorporates bootstrap-based uncertainty quantification for evaluating variability in counterfactual estimates. Bootstrap sampling is performed through repeated resampling with replacement, allowing the framework to estimate the stability of intervention effects under varying data distributions.

For each bootstrap iteration, a Random Forest model is trained using a resampled dataset, and the corresponding counterfactual effect is computed. Let e_i denote the estimated counterfactual effect obtained from the i^{th} bootstrap iteration. The bootstrap mean is computed as:

$$\mu_{boot} = \frac{1}{N} \sum_{i=1}^N e_i$$

where N denotes the total number of bootstrap iterations.

The bootstrap standard deviation is computed as:

$$\sigma_{boot} = \sqrt{\frac{1}{N} \sum_{i=1}^N (e_i - \mu_{boot})^2}$$

To estimate statistical confidence, a 95% confidence interval is computed using percentile-based estimation:

$$CI_{95\%} = [P_{2.5}, P_{97.5}]$$

where $P_{2.5}$ and $P_{97.5}$ denote the lower and upper percentile bounds of the bootstrap distribution, respectively.

In addition to uncertainty estimation, robustness validation is performed using refutation methods provided by the causal inference framework. Random common cause refutation and data subset refutation are used to evaluate the stability of estimated causal effects under perturbed conditions.

G. Domain-General Evaluation Pipeline

To evaluate cross-domain applicability, the proposed framework is extended to multiple heterogeneous datasets spanning educational, healthcare, and marketing domains. Each dataset is processed through a unified causal reasoning pipeline consisting of causal estimation, counterfactual simulation, and uncertainty quantification.

Preprocessing operations such as categorical encoding and treatment normalization are applied to ensure compatibility across datasets. In the marketing dataset, treatment variables are generated by converting email intervention categories into binary treatment assignments. The unified pipeline enables

consistent causal analysis despite differences in domain characteristics and feature distributions.

To evaluate agreement between causal estimation and uncertainty-aware simulation, a consistency metric is computed as:

$$Consistency = |ATE - \mu_{boot}|$$

This metric quantifies the deviation between the estimated average treatment effect and the bootstrap-based counterfactual mean. Lower consistency values indicate stronger agreement between structural causal estimation and simulation-based intervention analysis.

The proposed domain-general evaluation pipeline enables scalable causal reasoning across heterogeneous environments while preserving interpretability, robustness, and uncertainty-aware decision support.

IV. EXPERIMENTAL SETUP

This section describes the datasets, preprocessing procedures, implementation environment, and evaluation configuration used to assess the proposed unified structural counterfactual framework across heterogeneous domains.

A. Datasets

To evaluate the generalization capability of the proposed framework, experiments were conducted on three real-world datasets spanning educational, healthcare, and marketing domains.

1) *Student Performance Dataset*: The Student Performance dataset [9], [10] was used to evaluate causal reasoning within an educational setting. The dataset contains student-related attributes including study time, health condition, absences, failures, and academic performance indicators. In the proposed framework, study time was selected as the treatment variable, while the final grade ($G3$) was considered the outcome variable. Variables such as absences, health condition, and previous failures were treated as confounding factors during causal estimation.

2) *Pima Indians Diabetes Dataset*: The Pima Indians Diabetes dataset [11] was used to evaluate the framework in a healthcare setting. The dataset contains medical diagnostic measurements associated with diabetes prediction. Glucose level was selected as the treatment variable, while the diabetes outcome indicator was considered the outcome variable for causal analysis and counterfactual estimation.

3) *MineThatData Email Marketing Dataset*: The MineThatData email marketing dataset introduced by Kevin Hillstrom [12] was used to evaluate domain-general causal reasoning in a marketing environment. The dataset contains customer interaction and conversion information associated with email campaign interventions. Email exposure categories were transformed into binary treatment assignments to enable intervention-based analysis, while customer conversion was considered the outcome variable.

B. Data Preprocessing

Prior to causal analysis, preprocessing operations were performed to standardize heterogeneous datasets for unified pipeline execution. Missing value inspection and statistical analysis were conducted during the data understanding stage to verify dataset consistency.

Categorical attributes were transformed using one-hot encoding with redundant feature removal to ensure compatibility with machine learning models and causal estimation procedures. In the marketing dataset, treatment assignments were generated by converting email intervention categories into binary variables representing intervention and non-intervention conditions.

The preprocessing pipeline was designed to maintain structural consistency across domains while preserving dataset-specific causal characteristics.

C. Causal Modeling Configuration

The proposed framework utilized the DoWhy causal inference library [4] for causal effect identification and estimation. Directed acyclic graphs (DAGs) were generated programmatically using predefined treatment, outcome, and confounding relationships.

Causal effects were identified using the backdoor adjustment criterion and estimated through linear regression-based adjustment methods. For the educational dataset, the causal graph incorporated directed relationships between study time, student performance, and confounding variables including absences, health condition, and failures.

The average treatment effect (ATE) was estimated using:

$$ATE = \mathbb{E}[Y(1) - Y(0)]$$

which quantifies the expected intervention effect between treated and untreated conditions.

D. Counterfactual Simulation Setup

Counterfactual simulation was performed using the Random Forest Regressor model implemented through the scikit-learn framework. The model was trained using predictive feature sets defined within the schema configuration stage.

For each observational instance, intervention-based counterfactual samples were generated by modifying treatment variables while preserving remaining feature attributes. Original and counterfactual outcomes were then predicted using the trained regression model.

The counterfactual effect for each instance was computed as:

$$\Delta_{cf} = \hat{Y}_{cf} - \hat{Y}_{orig}$$

where $\hat{Y}_{cf}$ denotes the counterfactual prediction and $\hat{Y}_{orig}$ denotes the observational prediction.

To estimate dataset-level intervention effects, individual counterfactual effects were aggregated across all observational samples.

E. Uncertainty Quantification Setup

Bootstrap-based uncertainty estimation was incorporated to evaluate the stability of counterfactual predictions under varying data distributions. For the educational dataset, 100 bootstrap iterations were performed through repeated sampling with replacement.

For each bootstrap iteration, a new Random Forest model was trained using the resampled dataset, and the corresponding counterfactual effect was computed. The bootstrap mean and bootstrap standard deviation were computed using the corresponding statistical formulations for repeated counterfactual effect estimation.

$$\mu_{boot} = \frac{1}{N} \sum_{i=1}^N e_i$$

and

$$\sigma_{boot} = \sqrt{\frac{1}{N} \sum_{i=1}^N (e_i - \mu_{boot})^2}$$

A percentile-based confidence interval was further computed as:

$$CI_{95\%} = [P_{2.5}, P_{97.5}]$$

to evaluate uncertainty bounds associated with intervention effects.

F. Generalization and Evaluation Configuration

To evaluate cross-domain applicability, the proposed framework was executed independently across educational, healthcare, and marketing datasets using a unified causal reasoning pipeline. For domain-general evaluation, 30 bootstrap iterations were performed for each dataset during generalized counterfactual estimation.

Evaluation focused on four primary measurements:

- Average Treatment Effect (ATE)
- Counterfactual Effect
- Bootstrap Mean
- Bootstrap Standard Deviation

In addition, a consistency metric was computed to measure agreement between structural causal estimation and bootstrap-based counterfactual simulation:

$$Consistency = |ATE - \mu_{boot}|$$

Lower consistency values indicate stronger agreement between estimated causal effects and uncertainty-aware counterfactual simulation outcomes.

G. Implementation Environment

The proposed framework was implemented in Python using libraries including pandas, NumPy, scikit-learn, matplotlib, and DoWhy. Experimental execution was performed using Jupyter Notebook-based modular pipeline stages corresponding to data understanding, causal modeling, counterfactual simulation, uncertainty quantification, and evaluation workflows.

V. RESULTS AND DISCUSSION

This section presents the experimental results obtained from the proposed unified structural counterfactual framework across educational, healthcare, and marketing domains. The analysis focuses on causal estimation behavior, counterfactual intervention modeling, uncertainty quantification, and consistency evaluation under heterogeneous datasets.

A. Cross-Domain Causal Evaluation

The proposed framework was evaluated across three heterogeneous datasets representing educational, healthcare, and marketing environments. For each dataset, the framework computed the Average Treatment Effect (ATE), counterfactual intervention effect, bootstrap-based uncertainty statistics, confidence intervals, and consistency measurements between structural causal estimation and simulation-based intervention analysis.

Table I presents the overall causal evaluation and uncertainty estimation results across all evaluated domains.

The educational dataset produced the highest Average Treatment Effect among all evaluated domains, indicating that study time exhibited comparatively stronger causal influence on student performance. In contrast, the healthcare and marketing datasets produced smaller intervention effects, suggesting comparatively weaker observable causal relationships under the selected treatment configurations.

The counterfactual intervention estimates further demonstrated that the proposed framework successfully generated simulation-based intervention predictions across heterogeneous datasets while maintaining consistency between structural causal estimation and machine learning-based counterfactual reasoning.

B. ATE and Bootstrap Mean Comparison

To evaluate agreement between structural causal estimation and uncertainty-aware counterfactual simulation, the framework compared Average Treatment Effect (ATE) estimates with bootstrap-based counterfactual means across all domains.

Fig. 2 illustrates the comparison between estimated causal effects and bootstrap-based intervention estimates.

The educational dataset exhibited a comparatively large difference between the estimated ATE and the bootstrap mean. This observation indicates stronger variability and nonlinear intervention behavior during simulation-based counterfactual estimation within the educational domain. Although the bootstrap mean remained positive, the deviation from the structural causal estimate suggests increased sensitivity to observational variation and resampled data distributions.

In comparison, the healthcare and marketing datasets produced relatively smaller differences between estimated ATE and bootstrap-based intervention effects. These results indicate improved agreement between structural causal estimation and counterfactual simulation within domains exhibiting comparatively lower intervention variability.

TABLE I
CROSS-DOMAIN CAUSAL EVALUATION AND UNCERTAINTY ANALYSIS RESULTS

Domain	ATE	CF_effect	Bootstrap_mean	Bootstrap_std	CI_low	CI_high	Consistency
Education	0.534	0.01	0.0173	0.0564	-0.0391	0.0737	0.5167
Healthcare	0.007	0.00	-0.0023	0.0056	-0.0079	0.0033	0.0093
Marketing	0.005	0.00	0.0000	0.0000	0.0000	0.0000	0.0050

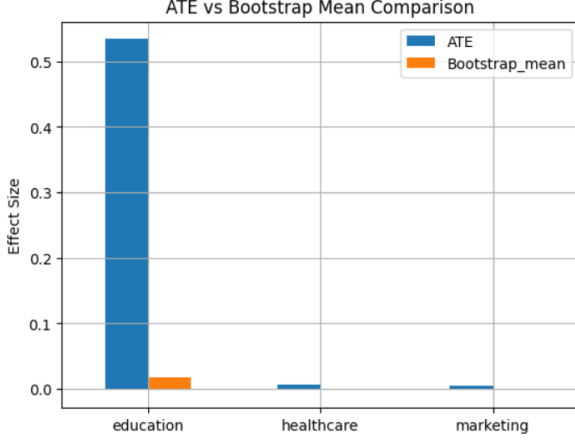


Fig. 2. Comparison between Average Treatment Effect (ATE) and bootstrap-based counterfactual mean across evaluated domains.

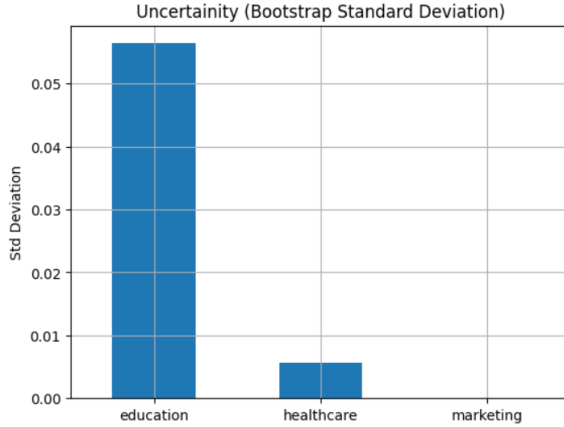


Fig. 3. Bootstrap standard deviation across educational, healthcare, and marketing datasets.

C. Bootstrap Uncertainty Analysis

Bootstrap-based uncertainty estimation was performed to evaluate the stability of counterfactual intervention effects under repeated resampling conditions. The bootstrap standard deviation was used as the primary uncertainty measurement for analyzing intervention variability across domains.

Fig. 3 presents the bootstrap standard deviation obtained across all evaluated datasets.

The educational dataset produced the highest bootstrap standard deviation, indicating greater uncertainty and variability in estimated intervention effects. This behavior may be attributed to stronger feature interactions, heterogeneous student characteristics, and varying observational distributions within the educational dataset.

The healthcare dataset demonstrated comparatively lower uncertainty, suggesting more stable intervention behavior under repeated bootstrap sampling. Similarly, the marketing dataset produced near-zero bootstrap variance, indicating highly stable but comparatively weak intervention effects under the evaluated treatment configuration.

The computed confidence intervals further supported these observations. The educational dataset exhibited wider confidence bounds relative to healthcare and marketing domains, confirming increased variability in intervention estimation. These findings demonstrate that the proposed framework effectively captures domain-specific uncertainty characteristics while preserving causal interpretability.

D. Consistency Evaluation

To quantitatively evaluate agreement between structural causal estimation and simulation-based intervention modeling, a consistency metric was computed as:

$$Consistency = |ATE - \mu_{boot}|$$

where ATE represents the estimated Average Treatment Effect and μ_{boot} denotes the bootstrap-based counterfactual mean.

Lower consistency values indicate stronger agreement between structural causal estimation and uncertainty-aware counterfactual simulation. The healthcare and marketing datasets produced relatively lower consistency deviations, indicating improved alignment between structural estimation and intervention simulation. In comparison, the educational dataset exhibited a larger consistency value due to increased bootstrap variability and stronger differences between estimated and simulated intervention effects.

Despite domain-specific variations, the proposed framework maintained stable causal reasoning behavior across heterogeneous environments without requiring domain-specific architectural redesign.

E. Discussion

Overall, the experimental results demonstrate that the proposed framework successfully integrates structural causal inference, counterfactual simulation, uncertainty quantification, and domain-general evaluation within a unified pipeline. The framework maintained interpretable causal reasoning behavior across heterogeneous datasets while explicitly modeling intervention uncertainty and simulation consistency.

Unlike conventional predictive systems that primarily focus on statistical correlations, the proposed approach enables intervention-aware reasoning through integration of structural

causal modeling, machine learning-based counterfactual estimation, bootstrap uncertainty analysis, and cross-domain evaluation.

The experimental findings further demonstrate that uncertainty characteristics vary substantially across domains, emphasizing the importance of integrating uncertainty quantification directly within causal reasoning pipelines. By combining causal estimation, counterfactual simulation, and uncertainty-aware evaluation within a scalable framework, the proposed system provides a reliable and interpretable solution for domain-general causal reasoning under uncertainty.

VI. ROBUSTNESS ANALYSIS

To evaluate the reliability and stability of the proposed unified structural counterfactual framework, robustness analysis was performed using refutation techniques provided by the DoWhy causal inference framework. The objective of this analysis was to determine whether the estimated causal effects remained stable under perturbed observational conditions and synthetic confounding variations.

The proposed framework utilized two robustness validation strategies:

- Random Common Cause Refutation
- Data Subset Refutation

These refutation methods were applied after causal effect estimation to assess sensitivity of the estimated intervention effects under varying causal and observational conditions.

A. Random Common Cause Refutation

The random common cause refuter evaluates robustness by introducing an additional synthetic confounding variable into the causal graph and recomputing the estimated causal effect. Stable causal estimates under this perturbation indicate that the framework is less sensitive to unobserved confounding influences.

The random common cause refutation produced the following results:

- Estimated effect: 0.20336046135752106
- New effect: 0.2033604613575247
- p-value: 0.0

The negligible difference between the original estimated effect and the refuted effect demonstrates that the proposed framework maintained highly stable causal estimation behavior even after introduction of synthetic confounding variables.

The observed stability indicates that the structural causal relationships modeled within the framework remain consistent under additional perturbation conditions. These findings suggest that the proposed causal estimation pipeline exhibits robustness against minor unobserved confounding variations.

B. Data Subset Refutation

The data subset refuter evaluates robustness by estimating causal effects using randomly sampled subsets of the original observational dataset. This analysis determines whether estimated intervention effects remain stable under partial observational perturbation and reduced sampling conditions.

The data subset refutation produced the following results:

- Estimated effect: 0.20336046135752106
- New effect: 0.19586555400910843
- p-value: 0.98

Although a small variation was observed between the original and subset-based estimates, the overall causal effect remained relatively stable under partial data perturbation. The high p-value further indicates that the observed deviation was not statistically significant under the evaluated conditions.

These observations demonstrate that the proposed framework maintains consistent causal reasoning behavior even when observational data distributions are partially altered through random subset sampling.

C. Robustness Interpretation

The robustness analysis demonstrates that the proposed unified structural counterfactual framework maintains stable causal estimation behavior under both synthetic confounding perturbation and partial observational sampling variations.

The random common cause analysis confirmed that estimated intervention effects were minimally affected by introduction of additional confounding variables, indicating structural stability of the causal estimation pipeline. Similarly, the data subset analysis demonstrated that the framework preserved causal consistency even under reduced observational conditions.

These findings indicate that the proposed framework provides reliable causal reasoning behavior across varying perturbation scenarios while maintaining interpretability and consistency of estimated intervention effects.

Overall, the robustness validation results support the reliability of the proposed framework for domain-general causal reasoning under uncertainty.

VII. LIMITATIONS

Although the proposed unified structural counterfactual framework demonstrates stable causal reasoning behavior across multiple heterogeneous domains, several limitations remain within the current implementation scope.

First, the framework relies on predefined causal structures for directed acyclic graph (DAG) construction. Treatment variables, outcome variables, and confounding relationships are manually specified during schema definition and graph generation stages. Consequently, the current framework does not perform automatic causal discovery from observational data and depends on prior structural assumptions regarding causal relationships.

Second, the implemented counterfactual simulation strategy primarily focuses on single-variable intervention analysis. Counterfactual samples are generated through direct modification of treatment variables while preserving remaining feature attributes. More complex multi-variable interventions and dynamic intervention policies were not explored within the current implementation.

Third, the proposed framework was evaluated using medium-scale observational datasets across educational,

healthcare, and marketing domains. Although the experimental results demonstrate cross-domain applicability, large-scale real-world deployment scenarios involving substantially larger observational distributions were not evaluated within the current study.

In addition, the causal estimation pipeline operates primarily on observational datasets and may still be influenced by unobserved confounding factors that are not explicitly represented within the constructed causal graphs. While robustness validation and refutation analysis demonstrated stable causal behavior under evaluated perturbation conditions, complete elimination of hidden confounding effects cannot be guaranteed in observational causal inference settings.

The uncertainty quantification stage utilized bootstrap-based estimation with fixed iteration counts for repeated sampling analysis. Although the implemented bootstrap strategy provided stable uncertainty measurements across evaluated domains, larger bootstrap sampling configurations may further improve statistical stability and confidence estimation under broader observational settings.

Finally, the proposed framework focuses primarily on static causal reasoning using observational snapshot datasets. Temporal causal dynamics, sequential intervention modeling, and time-dependent causal evolution were not incorporated within the current implementation.

Despite these limitations, the proposed framework establishes a scalable and interpretable foundation for integrating structural causal inference, counterfactual simulation, uncertainty quantification, and cross-domain reasoning within a unified causal analysis pipeline.

VIII. CONCLUSION

This paper presented a unified structural counterfactual framework for domain-general causal reasoning under uncertainty. The proposed framework systematically integrated data understanding, causal graph construction, structural causal effect estimation, machine learning-based counterfactual simulation, bootstrap-driven uncertainty quantification, robustness validation, and cross-domain evaluation within a single interpretable pipeline.

Unlike conventional predictive systems that primarily rely on statistical correlations, the proposed approach enabled intervention-aware reasoning through integration of structural causal inference and simulation-based counterfactual analysis. The framework further incorporated uncertainty estimation and refutation-based validation to improve reliability and interpretability of estimated causal effects under heterogeneous observational conditions.

Experimental evaluation was conducted across educational, healthcare, and marketing datasets to assess generalization capability under multiple domains. The results demonstrated that the proposed framework successfully maintained consistent causal reasoning behavior across heterogeneous environments while effectively capturing domain-specific intervention variability and uncertainty characteristics. The integration of bootstrap uncertainty estimation and consistency evaluation additionally provided deeper insight into stability of simulated intervention effects across varying observational distributions.

Robustness analysis further demonstrated that the estimated causal effects remained stable under synthetic confounding perturbation and partial observational sampling conditions. These findings indicate that the proposed framework provides reliable and interpretable causal reasoning behavior under uncertainty-aware evaluation settings.

Overall, the proposed unified structural counterfactual framework establishes a scalable foundation for integrating causal inference, counterfactual reasoning, uncertainty quantification, and domain-general evaluation within a cohesive analytical system. The framework contributes toward development of interpretable and uncertainty-aware causal reasoning pipelines capable of supporting real-world decision-making across heterogeneous application domains.

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